PeriCoupler Plugin-User Instructions

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1 Prerequisites

Before launching the plugin, ensure the following requirements are met:

- ABAQUS 6.14 or later is installed with a valid license.
- The PeriCoupler plugin files are placed in the ABAQUS plugin directory, typically located at: C:\SIMULIA\Abaqus\plugins\PD_Comp_P\
- No external Python packages required beyond the ABAQUS Python environment.
- No prior ABAQUS model database (.cae) is required; the plugin generates the model from scratch.

2 Launching the Plugin

1. Open ABAQUS/CAE.
2. From the top menu bar, navigate to: Plug-ins → PD Plate Plugin → Launch PD Plate Plugin.
3. The graphical user interface (GUI) will appear, presenting a structured input panel organized into four input categories as shown in figure 1.

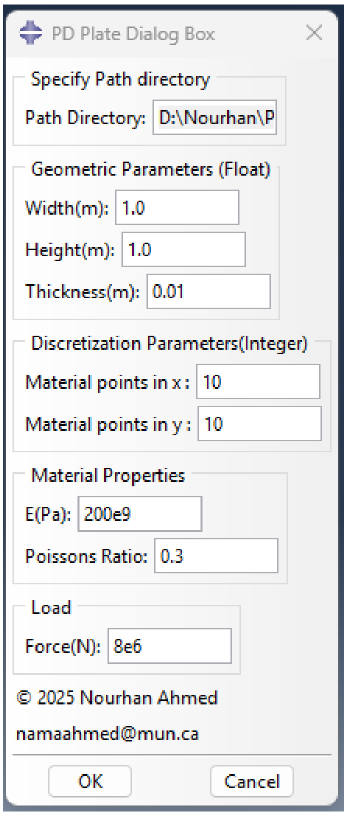

Figure 1: PeriCoupler Dialog Box

3 Defining User Inputs

The GUI panel is divided into the following input sections:

3.1 Specify Path Directory

Identify the work directory path.

3.2 Geometric Parameters

Enter the following dimensional quantities as float:

- Width (L_x): Total length of the domain in the x-direction in meters.
- Height (L_y): Total length of the domain in the y-direction in meters.
- Thickness (t): Out-of-plane thickness of the domain (used for volume calculations under plane stress assumptions) in meters.

3.3 Discretization Parameters (Integer)

- Material points in x: Number of uniformly spaced material points along the x-direction.
- Material points in y: Number of uniformly spaced material points along the y-direction.

3.4 Material Properties

- Young's Modulus (E): Elastic modulus of the material in Pa.
- Poisson's Ratio (ν): Used to compute the PD micro-modulus cc c for the bond-based formulation.

3.5 Loading Conditions

Enter the applied loading:

- Applied Force (F_x): Magnitude of the force applied in the x-direction.
- Boundary conditions and load assignments are automatically generated based on this input.

4 Model Generation Procedure

4.1 PD Mapping and Micro-modulus Evaluation

- The volume of each material point V_j is computed based on the grid spacing and thickness.
- The PD micro-modulus cc c is evaluated from the specified material properties.
- The equivalent PD cross-sectional area A_{PD} and elastic modulus E_{PD} are computed to map the PD bond stiffness onto the truss element properties.

Upon confirming the inputs, the plugin executes the following automated sequence:

4.2 Grid Point Generation

- A structured grid of material points is generated over the domain $L_x \times L_y$based on the specified discretization parameters.
- The grid spacing Δx and Δy are computed automatically from the domain dimensions and point counts.

4.3 Horizon Construction

- The horizon δ is computed as a fixed multiple of the grid spacing (typically $\delta = 3\Delta x$).
- For each material point, all neighboring points within the horizon radius are identified based on the Euclidean distance criterion: $\mid x_j - x_k \mid \leq \delta$

4.4 Material and Section Assignment

- A linear elastic material is created using the specified E and ν values with thethe equivalent PD elastic modulus E_{PD}.
- A truss cross-sectional area is defined and assigned to all bond elements with the equivalent PD cross-sectional area A_{PD}.

4.5 Element Generation

- For each identified bond, the two end nodes P_1 and P_2 are defined.
- A linear two-node truss element (T2D2 in ABAQUS) is created for each bond.
- This process iterates until all material point pairs within the horizon have been assigned a truss element as shown in figure 2.

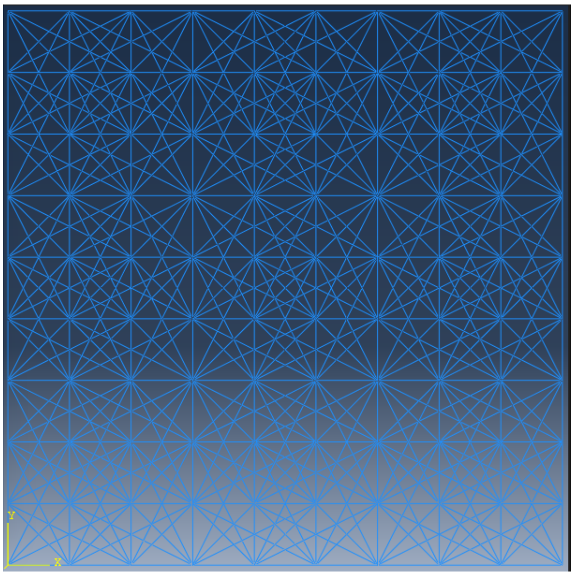

Figure 2: Element Generation

4.6 Boundary Conditions and Load Application

- Displacement boundary conditions are applied to the designated boundary material points.
- The applied force F_x is distributed across the loaded boundary nodes.

4.7 Mesh Construction

- The element type is assigned and the mesh is finalized within ABAQUS.
- No additional meshing operations are required, as the truss elements constitute the complete discretization.

4.8 Running the Analysis

- Once the model is fully generated, the job will be submitted automatically.
- No additional job operations are required.

4.9 Post-Processing and Visualization

Upon successful completion of the analysis, the following outputs are available in the Visualization module:

- Displacement Contour Map: Displays the nodal displacement field U over the deformed domain as illustrated in figure 3.
- PD Stress Contour Map: Displays the mapped peridynamic stress field computed from the bond stretch values using the strain mapping procedure as illustrated in figure 4.

To access these outputs:

- Open the Output Database (.odb) file from the Job Manager.
- Navigate to Results → Field Output.
- Select U for displacement or the mapped PD stress variable for the stress contour.

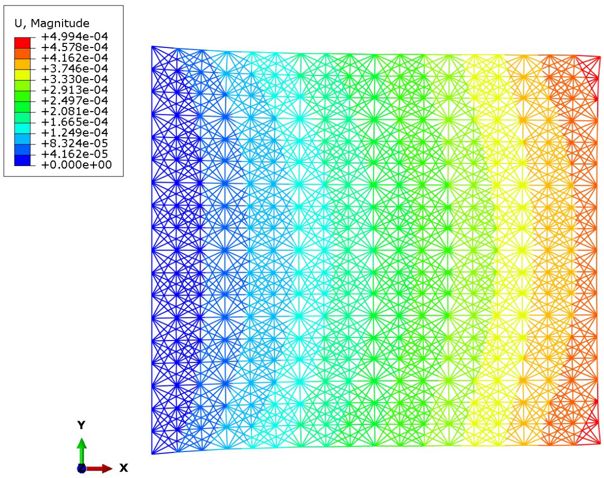

Figure 3: Displacement Contour map

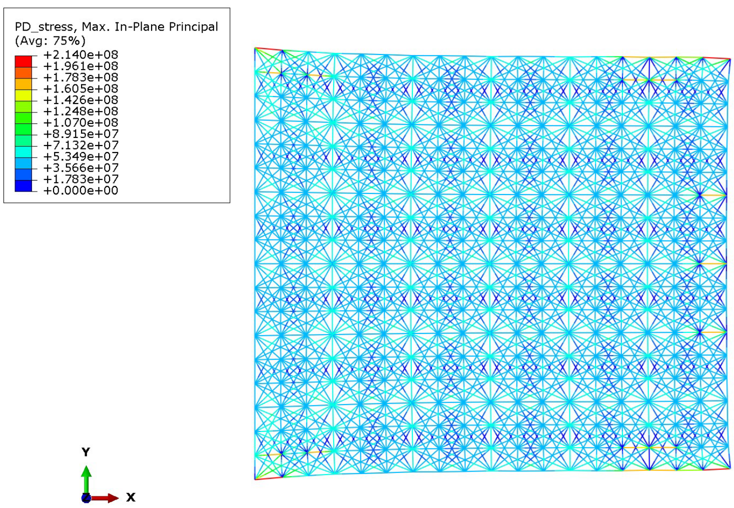

Figure 4: PD stress contour map